1.1 - Limits

Goals: Describe limits graphically, numerically, and algebraically.
Before Class Videos: 1.1.1 and 1.1.2

Example 1: Consider these three functions:

$$f(x) = x + 1, \qquad g(x) = \frac{x^2 - 1}{x - 1} = \frac{(x+1)(x-1)}{x-1}, \qquad h(x) = \begin{cases} x + 1, & x \neq 1, \\ 3, & x = 1. \end{cases}$$

How are these similar or different?

Definition:

$\lim_{x \to a} f(x) = L$ if $f(x)$ can be made arbitrarily close to L for all values of x sufficiently close to a.

Example 2: Compute the limit as $x \to 1$ of $f(x)$, $g(x)$, and $h(x)$.

Example 3: Use a table of values to estimate $\lim_{x \to 1}(x + 1)$.

Example 4: Use a table of values to estimate $\lim_{x \to 0} \frac{1}{x}$.

Example 5: Investigate

$$f(x) = \begin{cases} x, & x < 1, \\ x + 1, & x \geq 1. \end{cases}$$

Definition:

$\lim_{x \to a^+} f(x)$ means $f(x)$ gets arbitrarily close to L for all x sufficiently close to a from the right, i.e. $x > a$.

Example 6: Investigate limits of

$$f(x) = \sin\left(\frac{1}{x}\right)$$

Example 7: Graphically investigate limits of $\frac{x^2-1}{x^2-x-2}$

Suggested after class problems from APEX: Section 1.1 problems 9,11,15